Enhancing Decision-Making: The Impact of Information Provision on Worker Task Assignment

Yunzhu Lai*

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Abstract

Managers often face substantial challenges in assigning the right workers to the right tasks, in part because they have limited information about workers' relative skills and preferences. We conduct a randomized field experiment in three large garment manufacturing plants in China to study whether providing managers with information about workers improves task allocation and firm outcomes. We provide randomly selected team managers with reports on workers' relative efficiency across tasks, including overall efficiency, learning speed, efficiency variability by task complexity, and performance in multi-worker versus solo tasks; for a subset of teams, the reports additionally include workers' task preferences. We find that providing information increases team efficiency by approximately 10 percent and reduces worker turnover within four months by 34–47 percent. Exploring the underlying mechanisms, we find that the intervention substantially improves managers' knowledge of workers. However, using detailed task-level administrative data, we find that managers incorporate only workers' relative overall efficiency into their assignment decisions, with no corresponding evidence that they adjust assignments based on the other dimensions of worker performance or preferences. The results show that information provision can improve worker-task matching and organizational performance, while highlighting a gap between managers' ability to acquire information and their ability to apply multiple dimensions of that information in complex assignment decisions.

Keywords: Task Allocation, Information Friction, Firm Performance

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*Agricultural and Resource Economics, University of Maryland. Email: ylai39@umd.edu.

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1 Introduction

Management practices are an important determinant of firm performance, and firms in developing countries tend to have weaker management practices than firms in developed countries (Bloom and Van Reenen, 2010; Bloom et al., 2013; Cai and Wang, 2022). One important dimension of management is how managers allocate workers across tasks. The way workers are matched to tasks can have substantial consequences for both productivity and worker retention (Adhvaryu, Kala and Nyshadham, 2022; Bergeron et al., 2022; Chen et al., 2025). Yet assigning the right worker to the right task is often difficult. In manufacturing plants, managers may need to assign dozens of workers across many tasks within a short period of time, while facing tight production deadlines and limited information about workers' relative skills and productivity in different tasks. These constraints may be particularly important for firms in developing-countries, where managers often have limited access to formal management training and decision-support tools.

One potentially scalable way to assist managers is to provide information on workers' performance across tasks, summarized from historical production data. Information frictions can prevent managers from accurately assessing workers' comparative strengths and assigning them to the tasks where they are most productive. Existing literature has examined various mechanisms contributing to worker-task misallocation, including manager inattentiveness (Adhvaryu, Kala and Nyshadham, 2022), insufficient team-level incentives (Burgess et al., 2010), and informal relationships between workers and managers (Bandiera, Barankay and Rasul, 2009). However, we know less about whether providing managers with systematic information about worker productivity can improve task allocation, which dimensions of worker information are most useful, and whether managers actually incorporate newly available information into their decisions.

This paper provides experimental evidence on how information provision affects worker and firm outcomes, and whether the changes are induced by improvement in managers' knowledge and task-allocation decisions. We conduct a randomized field

experiment in three large-scale garment manufacturing plants in China, involving 42 sewing teams and 1,256 workers. The garment industry provides a particularly useful setting for studying this question. When plants introduce a new garment style, team managers must reassign roughly 20–40 workers across 30–100 or more tasks under tight delivery deadlines. Different styles require different combinations of machines and tasks, and workers vary substantially in their productivity across tasks. Baseline survey evidence indicates that these decisions are difficult for managers: 78.6 percent report management problems related to task allocation, and 81 percent report difficulty identifying workers' relative efficiency and skill specialization. Managers' limited knowledge is also evident in an objective knowledge test. We compare managers' pre-intervention assessments of the five most efficient workers on the flatbed machine and five most variable workers with rankings constructed from detailed worker-task production records. On average, managers correctly identify only 2.7 of the five workers with the highest measured efficiency and 0.8 of the five with the highest measured efficiency variability.

In our experiment, we randomly assign teams to an efficiency information treatment, an efficiency and preference information treatment, or a control group. In both treatment arms, managers receive reports on individual worker efficiency constructed from administrative production records covering the year preceding the intervention. The reports summarize workers' relative efficiency across machines and task complexity levels, learning speed when beginning a new production cycle, variability in efficiency across task complexity levels, and comparative performance on multi-worker versus solo tasks. In the second treatment arm, managers additionally receive information on workers' preferences over machines, task complexity, and garment parts, elicited through a baseline worker survey. The intervention is designed not only to provide information but also to facilitate its use: supervisors of team managers organized meetings to introduce the importance of the reports and their intended use, and additional individual meetings and follow-ups were carried out to to explain the measures and how they can inform task assignment.

We combine detailed administrative production data with baseline and follow-up surveys of workers and managers. The administrative data contain daily worker-task output and time-use records, task characteristics including machine type and complexity, monthly team-level efficiency and quality measures, worker rosters, and worker turnover. These data allow us to construct measures of worker productivity across tasks and to directly measure the quality of managers' task assignments. We administered baseline and follow-up surveys to test managers' knowledge of workers' skills and preferences. We also carried out both baseline and follow-up surveys to managers and workers to measures job satisfaction and evaluations of team managers.

We first document substantial effects of the intervention on team performance and worker retention. Team efficiency increases by 10.7 percent under the efficiency information treatment and 10.2 percent under the efficiency and preference information treatment. We find no detectable effect on product quality, suggesting that the gains in productivity do not come at the expense of quality. Worker turnover within four months falls by approximately 47 percent and 34 percent in the two treatment arms, respectively. These results show that providing managers with information about their workers can generate meaningful improvements in both team performance and worker retention.

We next examine the mechanisms underlying these effects. One month after the intervention, both treatment arms substantially increase managers' knowledge of workers' relative efficiency, learning speed, efficiency variability, and relative performance on multi-worker versus solo tasks, with knowledge gains of roughly 0.7–0.9 standard deviations across these dimensions. There is no statistically detectable difference between the two treatments. The efficiency and preference information treatment also increases managers' knowledge of worker preferences by 1.34 standard deviations, whereas the efficiency information treatment has little effect on preference knowledge. Especially, we find these knowledge gains mostly concentrates among team managers with relative weaker pre-intervention management practices. These gains in efficiency knowledge translate into changes in managers' task-allocation decisions. Using administrative

data, we construct a demand-adjusted measure of assignment quality that benchmarks realized assignments against the best feasible assignment given the team's task demand and available qualified workers. Both treatments improve assignment quality along the efficiency dimension by a 13-14 percent reduction in the shortfall from the best-feasible benchmark. Consistent with the heterogeneity in knowledge gains, we find larger effect on task assignment improvement among managers with less prior knowledge of workers' relative efficiency. Thus, the intervention not only improves managers' knowledge of workers but also leads them to assign workers more effectively across tasks, providing a mechanism for the observed gains in team efficiency and worker retention.

At the same time, the pattern of results reveals that managers selectively use the information they acquire. Although the intervention improves managers' knowledge across multiple dimensions—including efficiency, learning speed, performance variability, teamwork, and, when provided, worker preferences—we find little evidence that dimensions other than efficiency affect task-allocation decisions. In particular, the substantial increase in knowledge of worker preferences under the efficiency and preference information treatment does not translate into a detectable improvement in preference-based assignment or overall job satisfaction. Taken together, these findings suggest that the relevant constraint is not simply managers' access to information: managers may prioritize information they view as more important for meeting production objectives and may be reluctant to incorporate other dimensions when doing so makes task assignment more difficult or creates a perceived trade-off with team efficiency. The results therefore highlight an important distinction between making information available to managers and inducing them to incorporate that information into organizational decisions.

Our paper first relates to a large literature documenting the importance of management practices for firm performance. Management quality is strongly associated with productivity and other measures of firm performance (Bloom and Van Reenen, 2007), and experimental evidence shows that interventions that improve management prac-

tices can increase productivity and other firm outcomes in developing-country settings (Bloom et al., 2013; Bruhn, Karlan and Schoar, 2018; Cai and Wang, 2022). Our paper complements this literature by focusing on a specific managerial function—allocating workers to tasks. We provide experimental evidence on whether reducing managers' information frictions about worker skills improves task allocation and organizational performance. Existing research shows that worker-task matching matter for firm productivity. For example, Adhvaryu, Kala and Nyshadham (2022) show that managers in a garment manufacturing firm reallocate workers toward tasks in which they perform relatively well, which mitigate the negative productivity shocks. More recently, Minni (2026) provides evidence from a large multinational firm that managers affect workers' productivity and career progression in part through their ability to match workers to jobs within the firm. However, there is less evidence on whether managers' limited information about heterogeneous worker productivity is itself an important constraint on allocation. Our experiment directly changes the information available to managers, while detailed administrative and survey data allow us to trace a direct link from information provision to managerial knowledge, task-allocation decisions, and downstream firm and worker outcomes, rather than inferring changes in managerial behavior solely from final performance outcomes.

Our paper also contributes to a growing literature on the use of data and predictive technologies in managerial decision-making. As firms increasingly collect detailed data on their operations and workers, quantitative information and predictive tools have become increasingly important inputs into managerial decisions (Brynjolfsson and McElheran, 2016). A growing literature examines whether such tools can improve decisions traditionally made using managerial judgment. Chalfin et al. (2016) and Kleinberg et al. (2018), for example, demonstrate the potential for machine-learning predictions to improve human decision-making, while Hoffman, Kahn and Li (2018) shows that managers frequently override recommendations from job-testing technologies and that these overrides result in worse hires on average. Related field evidence shows that the benefits of data-driven decision tools depend critically on whether and how decision

makers incorporate their recommendations (Kesavan and Kushwaha, 2020; Kawaguchi, 2021). We contribute to this literature by studying how managers respond to multiple dimensions of worker information that may be relevant for task assignment. While managers may seek to assign workers to tasks where they are most productive, workers may also value particular machines, task difficulty levels, or parts of the production process. Such preferences constitute an economically meaningful dimension of the assignment problem: existing literature shows that workers place substantial value on non-wage job attributes (Mas and Pallais, 2017; Wiswall and Zafar, 2018; Maestas et al., 2023), and giving workers greater autonomy over task assignment can affect worker behavior and satisfaction with the assignment process (Kwasnitschka et al., 2026). Our experimental design therefore varies whether managers receive information on workers' productivity alone or information on both productivity and workers' stated task preferences. By measuring both managers' subsequent knowledge and their actual task assignments, we can examine not only whether managers learn about these different dimensions of worker characteristics, but also whether they incorporate them into their assignment decisions. This allows us to distinguish managers' acquisition of information from their use of that information in decision-making. We find that managers learn substantially about the worker characteristics provided to them, but selectively incorporate this information into their decisions: they improve worker-task matching along the efficiency dimension while making little detectable change in assignments along other dimensions, including worker preferences. Our findings therefore highlight that expanding the information available to managers need not translate into corresponding changes in managerial decisions across all dimensions.

The article proceeds as follows. Section 2 describes the institutional setting and the task-allocation problem in the garment manufacturing plants. Section 3 presents the experimental design and the information reports provided to managers. Section 4 describes the administrative and survey data. Section 5 presents the effects of the intervention on worker and firm outcomes and underlying mechanism. Section 6 concludes.

2 Background

2.1 Study Context

We collaborate with three garment manufacturing plants in China to study how information provision can improve managers' task allocation decisions and firm outcomes. The garment industry offers a particularly suitable setting for this research agenda.

Garment manufacturing plants of the type we study are widespread across the developing world. The production processes and organizational structure we describe below are common to garment factories throughout China, Bangladesh, Vietnam, India, and other major garment exporting countries. The textile and garment sector employs around 94 million workers globally (International Labour Organization, 2023). This type of labor-intensive manufacturing absorbs a large share of the low-skilled workforce in developing countries and constitutes a key entry point into formal employment, particularly for women and rural-to-urban migrants (Bárcia de Mattos et al., 2022; International Labour Organization, 2023). Improvements in how these plants allocate and manage labor therefore have direct implications not only for firm productivity but also for the earnings and welfare of a large population of workers.

Moreover, a large literature documents that management practices in developing-country manufacturing firms often fall short of best practices (Bloom and Van Reenen, 2007, 2010; Bloom et al., 2013), and the plants in our sample share many of these features. Over 78% of team managers in our sample plants have education level of junior high school or below, and have served as managers for 4.2 years on average. Almost all of them were promoted from the ranks of experienced sewing workers, and plants don't provide any formal management training. As a result, management decisions—including the allocation of workers to tasks—are made largely on the basis of personal experience and intuition rather than systematic information or formal decision-making tools.

2.2 The Sample Plants

The three plants in our sample are highly similar in their organization and production processes. Each plant receives production orders from apparel retailers, purchases garment materials from upstream textile firms, and produces a variety of garment styles. Production within each plant is organized into three workshops: cutting, sewing, and finishing. Fabric is first cut into garment parts in the cutting workshop; these parts are then stitched into finished garments in the sewing workshop; finally, garments are inspected, ironed, and packaged in the finishing workshop.

We focus on the sewing workshop, which is the most labor-intensive stage of production and employs the majority of each plant's production workers. Sewing workers operate a variety of sewing machines (e.g., flat sewing machine, covering stitch machine, overlock machine, chain stitch machine and so on) and perform a wide range of tasks that differ in required techniques and difficulty. Workers are organized into sewing teams—42 teams across the three plants in our sample—with each team comprising around 30 workers on average. Each team functions as an independent production unit: it produces a garment product, such as a T-shirt or a jacket, by sequentially stitching cut garment parts into a finished product, with each worker on the team responsible for one or several tasks along the assembly sequence. Sewing workers' monthly earnings are determined by task-specific piece rates and output volume. Team managers' monthly compensation consists of a fixed salary plus a bonus tied to team performance—specifically, team-level efficiency and quality.[1]

2.3 Style Changeovers and the Task Allocation Problem

A central feature of production in these plants is the frequent introduction of new garment styles. Because different styles require distinct sets of tasks and machinery, each style changeover requires the team manager to reassign workers to tasks. This reassignment problem is nontrivial: the manager must match roughly 20-40 workers to

[1]Managers' bonuses reflect team-level efficiency and quality, though the exact formula varies across plants: at one plant, the bonus depends on team efficiency, quality, and the number of style changeovers; at a second, it is based on the team's monthly output value; and at a third, it depends on team efficiency and quality alone.

a new configuration of tasks under tight delivery deadlines, with limited information about how productive each worker would be at each of the new tasks. Team managers in our sample expressed considerable difficulty in determining the optimal match between workers and tasks during changeovers. Around 81% of team managers in our sample reported in the baseline survey that identifying workers' expertise and skills is challenging.

Our discussions with managers and workers in these plants indicate that, as in manufacturing firms in other industries and countries, the style changeover period is critical and vulnerable, and significantly impacts profitability. Output drops sharply at the beginning of each changeover and recovers gradually as workers learn their newly assigned tasks. Any inefficiency in task allocation during this period can therefore lead to delayed deliveries, increased costs, and forgone profits.

The frequent style changes at these garment manufacturers, combined with the information frictions managers face, provide an ideal context to study the impact of information provision on managers' task allocation decisions and its downstream effects on team productivity and worker welfare.

3 The Experimental Design

To examine whether providing managers with information on worker efficiency and task preferences improves managers' knowledge of their workers, changes task allocation decisions, and ultimately affects team efficiency, job satisfaction, and worker retention, we conducted a randomized field experiment in three garment plants in China. Our sample comprises 42 sewing teams and 1,256 sewing workers. Teams were randomly assigned to one of three experimental groups: teams in the first treatment arm received reports on worker efficiency; teams in the second treatment arm received reports on both worker efficiency and workers' task preferences; and teams in the control group received no reports.

3.1 *The Efficiency and Preference Reports*

Prior to the intervention, we held meetings with plant directors, workshop managers, team managers, and industrial engineers at each plant to identify the metrics that managers considered most useful for allocating tasks and that the plants wished us to compile.

We constructed the efficiency reports using administrative production records covering the year preceding the intervention.[2] From these data, we computed measures of worker efficiency along four dimensions: (i) each worker's within-team efficiency ranking by machine type and task complexity level; (ii) each worker's within-team ranking in the rate of learning during the first five days of a new production cycle, when teams begin producing a new garment style; (iii) each worker's within-team ranking in the variability of efficiency by task complexity; and (iv) each worker's comparative advantage in multi-worker tasks relative to solo tasks.[3]

We constructed the preference reports using responses to the baseline worker survey, in which each worker was asked to select one to three preferred machines, one or two preferred task complexity levels, and one to three preferred garment parts, and to explain the reasons for these choices. Notably, very few managers in these plants had ever asked workers about their task preferences. Workers, in turn, were unwilling to communicate their preferences to managers absent some form of external facilitation. Many workers reported that they refrained from raising their preferences with their managers because they worries that doing so would be seen as resisting task assignments, or that it would impose an additional burden on management.

Reports were shared only with treated teams. The delivery of reports proceeded in two stages. First, each plant organized a formal introductory meeting for treated team managers. At this meeting, a senior plant manager introduced the report-sharing

[2]Output data are automatically logged by machines each time a worker completes one piece of a task. The data include daily output by worker and task, the total seconds a worker takes to complete each task, a description of the task, the standard allowable minutes (SAM) of the task, the machine required to perform the task, and the complexity level of the task.

[3]All four dimensions were recommended by team managers and industrial engineers during our pre-intervention meetings.

program, explained why the reports could be useful for task allocation, and provided concrete examples of how managers might use them. The research team then presented the content of the reports with illustrative slides, explained how each metric was computed, and described how the metrics could inform task allocation. Second, over the following weeks, the research team met individually with each treated team manager to walk through that team's report, explain how to interpret and apply each metric, and answer questions. Each individual session lasted approximately 40 minutes.

One month after the reports were delivered, we followed up with treated team managers regarding their use of the reports. To measure whether the reports updated managers' information about their workers, we also administered an in-person knowledge test to both treated and control team managers, in which managers answered questions about their workers' efficiency and task preferences without consulting the reports.

3.2 Randomization

We randomized treatment at the team level, stratifying by team size and past team performance. Team size is measured as the number of permanent workers on the pre-intervention roster. Past performance is measured using the team manager's average monthly performance bonus over the eight months preceding the intervention. For each stratification variable, we created an indicator equal to one if the team was above the within-plant sample median. Crossing the two indicators yields four strata within each plant. Within each stratum, teams were randomly assigned to one of the three experimental groups, resulting in 13 teams (358 workers) in treatment arm 1, which received only efficiency reports; 15 teams (370 workers) in treatment arm 2, which received both efficiency and preference reports; and 14 teams (402 workers) in the control group. Figure 1 illustrates this randomization explicitly.

[Insert Figure 1 Here]

3.3 Timing

Because preparing the reports required substantial data processing, the intervention was rolled out sequentially across the three plants. The first wave began in October 2025 in the first plant, with five teams assigned to the efficiency arm and four teams assigned to the efficiency-and-preference arm. The second wave began in February 2026 in the second plant, with two teams in the efficiency arm and three teams in the efficiency-and-preference arm. The third wave began in April 2026 in the third plant, with six teams in the efficiency arm and seven teams in the efficiency-and-preference arm.

Figure 2 shows the timeline for each individual plant. In Phase 1, we administered a baseline survey, held discussions with plant managers, and compiled reports for all teams simultaneously. Phase 2 constituted the intervention itself, beginning with an introductory meeting and followed by individual meetings with participants, as described above. One month after the intervention, Phase 3 involved manager follow-up together with a knowledge test to assess managers' understanding of the material covered. Finally, two months after Phase 3, Phase 4 consisted of a follow-up survey to evaluate the intervention's effects over time.

[Insert Figure 2 Here]

4 Data

Our analysis combines administrative data from the three plants with survey data collected from workers and team managers at baseline and follow-up between September 2025 and July 2026.

4.1 Administrative Data

We obtain three main sources of administrative data from the three plants. First, we receive daily worker-level production data. The production lines automatically record each worker's cumulative output and time spent on each assigned task in real time. The

data also contain information on task descriptions and the production line on which each worker performs the task. In addition, the industrial engineering departments provide information on task characteristics, including the required machine, complexity level,[4] and standard allowable time (SAT) in seconds.[5] We merge the daily production and task-characteristics data to calculate worker-level daily efficiency as $\frac{SAT \times \text{daily output}}{\text{time spent on the task}}$.

Second, we obtain monthly team-level performance data. Specifically, the plants provide monthly team efficiency and qualification rates calculated by their HR departments. The three plants use somewhat different measures of monthly efficiency. In two plants, covering 22 teams, monthly efficiency is calculated as $\frac{\sum_o N_o \times SAT_o}{\#\text{workers} \times \text{work hours per day} \times \text{work days per month}}$, where N_o and SAT_o denote the monthly output of task o and its standard allowable time, respectively. In the third plant, covering 20 teams, monthly efficiency is calculated as realized efficiency divided by target efficiency. Qualification rates are measured consistently across the three plants. Each garment is inspected before packaging, and the qualification rate is the share of inspected products that pass the quality check.

Third, we obtain worker roster data collected once before the intervention and again four months after the intervention. The rosters record current sewing workers, their team and position, as well as demographic characteristics including gender, age, and education, and their year of joining the company. We also obtain worker turnover records, including the date on which workers left the company, for the 8 months preceding the intervention and 4 months post intervention.

Table 1, Panel A reports summary statistics for variables constructed from the administrative data. The first two sets of variables describe demographic characteristics of managers and workers in the sewing departments. Managers and workers are both in their early 40s on average. Approximately 60% of team managers and 90% of

[4]The plants classify tasks into five complexity levels, from A to E, with A denoting the most complex tasks and E the simplest.

[5]SATs are drawn from the General Sewing Data (GSD), a widely used database in the garment manufacturing industry. Industrial engineers decompose each task into a sequence of basic movements, such as grabbing and aligning the garment, sewing a specified length, and turning the garment. The GSD assigns a standard time to each basic movement when performed by a qualified worker using the standard machine. Worker piece rates are typically determined based on the SAT and complexity level of the task.

workers are female. Managers have approximately four years of tenure at the plant on average, compared with three years for workers. Both managers and workers have approximately nine years of schooling on average, corresponding roughly to a junior-high-school education.

The sewing teams in our sample have approximately 30 workers on average. The average worker turnover rate during the eight months preceding the intervention is 15.2%. Average team efficiency and qualification rates over the six months preceding the intervention are 86.6% and 86.3%, respectively. Managers receive approximately 922 RMB performance bonus per month on average during the six months preceding the intervention. We find no statistically significant differences across the three treatment arms in these administrative characteristics.

4.2 Survey Data

We collect baseline surveys from workers and team managers prior to the intervention, sequentially across the three plants in August 2025, December 2025, and February 2026. We conduct follow-up surveys approximately three months after the intervention, in January 2026, May 2026, and July 2026.

Table 1, Panel B reports summary statistics for worker job satisfaction and evaluations of team managers. The worker survey includes questions on several dimensions of job satisfaction, measured on a five-point scale, with some questions phrased positively and others negatively. Several questions directly concern task allocation, including whether workers are satisfied with the tasks assigned to them during the preceding three months. Other questions capture workers' experiences with their team managers, including whether managers care about workers' feelings, treat workers fairly, whether workers would like to transfer to another team, and whether workers are frequently criticized by their managers. The survey also includes broader measures of satisfaction with the company, such as satisfaction with income during the preceding three months and whether workers have considered resigning from the company. Appendix Table A1.1 provides the complete list of questions. The average worker satisfaction score is

approximately 3.3 out of 5.

We also ask workers to evaluate their team managers along nine dimensions: organizing production, handling emergencies and conflicts, treating workers fairly, listening to workers' opinions and suggestions, caring about workers' work and personal lives, clearly communicating task requirements, providing ongoing guidance, encouraging workers during challenges, and being trustworthy. Each item is measured on a five-point scale. Appendix Table A1.2 provides the complete list of evaluation questions. The average manager evaluation score is approximately 3.7 out of 5.

[Insert Table 1 Here]

4.3 Survey Evidence on Managers' Knowledge

Baseline survey evidence indicates that team managers face substantial information and management constraints in allocating workers across tasks. When asked about the management problems they had encountered during the preceding three months, 78.6 percent of team managers reported challenges directly related to task allocation, including missed production deadlines, quality shortfalls, and assigning workers to tasks for which they were insufficiently qualified. Consistent with these difficulties, 81 percent of team managers reported that identifying workers' relative efficiency and skill specialization was challenging.

These self-reported difficulties are corroborated by an objective measure of managers' knowledge about worker productivity, collected prior to the intervention. We asked each team manager to identify the five workers on their team whom they believed to be the most efficient on the flatbed sewing machine—the most widely used machine on the factory floor—and, separately, the five workers whom they believed to exhibit the greatest volatility in efficiency. We then compared these nominations with workers' actual daily output records over the preceding year, which allow us to construct objective rankings of workers by mean efficiency and efficiency volatility. Table 2 reports managers' scores, defined as the number of correct matches, out of five,

between their nominations and the corresponding objective rankings. On average, team managers correctly identify only 2.7 of the five most efficient workers and 0.8 of the five workers with the greatest efficiency volatility. The substantial gap between managers' beliefs and workers' realized performance points to a meaningful information friction in managers' knowledge of worker efficiency.

Complementary evidence suggests that team managers rarely incorporate worker preferences into task-assignment decisions. In a multiple-response question regarding the criteria used in task assignment, 97.6 percent of team managers reported considering productivity-related factors, including skill, learning ability, and work attitude. By contrast, only 42 percent reported considering worker preferences, while the other 57 percent stated that preferences played no role in their assignment decisions. Moreover, even among managers who reported taking worker preferences into account, information is typically acquired indirectly rather than through direct communication: 66.7 percent reported inferring preferences from their observations of workers' behavior on the production line, whereas only 33 percent reported eliciting preferences directly from workers.

This limited consideration of worker preferences coincides with substantial challenges in workforce management. Over the same three-month period, 78.6 percent of team managers reported difficulties related to worker turnover, low motivation, or low job satisfaction.

[Insert Table 2 Here]

5 Main Results

5.1 Estimation Strategy

Teams were randomly assigned within randomization strata to one of three experimental arms: a control group, an efficiency-information treatment (T1), or an efficiency-and-preference-information treatment (T2). For each outcome, we estimate

an ANCOVA specification of the form

$$Y_{it} = \alpha + \beta_1 T1_i + \beta_2 T2_i + \rho, \overline{Y} * i^{\text{pre}} + \delta * s + \gamma_t + \varepsilon_{it}, \quad (1)$$

where i indexes the unit of observation and t indexes the post-intervention period. For outcomes measured repeatedly after the intervention—weekly team efficiency, monthly team quality, and daily task-assignment match quality—i indexes teams and t indexes the corresponding calendar period. In these specifications, $\overline{Y}_i^{\text{pre}}$ denotes team i's mean value of the same outcome over the pre-intervention periods, the term δ_s denotes randomization batch $\times$ stratum fixed effects, which account for the stratified randomization design, and γ_t denotes plant $\times$ calendar-time fixed effects, which absorb common shocks in outcomes across plants and over time.

For outcomes measured once after the intervention—worker turnover, satisfaction, team-level task assignment quality, and the knowledge test administered to team managers—t indexes the single post-intervention observation, and the period fixed effect γ_t is omitted. In these specifications, $\overline{Y}_i^{\text{pre}}$ is replaced by Y_i^{base}, the corresponding individual-level baseline measure.

The coefficients β_1 and β_2 capture the effects of the T1 and T2 treatments, respectively, relative to the control group, conditional on the corresponding baseline outcome. We additionally test the equality $\beta_1 = \beta_2$ to assess whether providing information about worker preferences in T2 generates effects beyond those of the efficiency information provided in both treatment arms. Standard errors are clustered at the team level, corresponding to the unit of randomization.

5.2 Results on Efficiency, Quality, and Turnover

We first examine the effects of the information intervention on team-level performance. Table 3 reports estimates from the ANCOVA specifications described above. We find that providing managers with information about worker efficiency increases team efficiency. In Column (1), the efficiency information treatment increases raw weekly team efficiency by 9.8 percentage points, relative to a control mean of 58.9%, correspond-

ing to an increase of approximately 17 percent. Providing both efficiency and preference information increases raw efficiency by 6.2 percentage points, although the estimate is somewhat less precisely estimated.

Raw efficiency may partly reflect differences in the complexity of the products assigned to teams. In particular, efficiency is mechanically more difficult to achieve for teams producing more complex styles, for which workers are required to perform more tasks simultaneously and face higher performance standards. To account for these differences, Column (2) reports results using efficiency adjusted for the complexity of the styles performed by each team. Complexity-adjusted efficiency nets out task complexity, measured as the number of distinct operations in a team's style. To avoid potential treatment effects, we estimate the coefficients on the number of operations using pre-intervention team-days only and then subtract this component from raw efficiency for each post-intervention team-day.[6] We provide details of the calculation of complexity-adjusted team efficiency in Appendix A.2. The efficiency information treatment increases complexity-adjusted efficiency by 8.5 percentage points, corresponding to a 10.7% increase relative to the control mean. The treatment providing both efficiency and preference information increases complexity-adjusted efficiency by 8.1 percentage points, or 10.2% relative to the control mean. The estimated effects of the two treatments are therefore very similar in magnitude. We cannot reject equality of the Treatment 1 and Treatment 2 effects for either measure of efficiency, with p-values of 0.373 and 0.906, respectively.

In contrast, we find no evidence that the intervention affects team quality. The estimated effects on monthly team quality are small and statistically insignificant for both Treatment 1 and Treatment 2. The point estimates are -0.007 and 0.001, respectively, relative to a control mean of 0.835, and we cannot reject equality of the two treatment effects. This result is consistent with the design of the intervention, which did not provide managers with information about worker performance on quality. Taken

[6]Appendix Table A1.4 shows that 1 additional task is associated with around 0.2% decrease in team efficiency, after controlling for learning speed.

together, these results indicate that providing managers with information about worker efficiency improves team efficiency without detectable effects on output quality.

[Insert Table 3 Here]

We next examine whether the information intervention affects worker turnover. Table 4 reports estimates from the ANCOVA specification on turnover effect. The sample includes workers who were still employed at the time of intervention, and the outcome is an indicator for whether a worker leaves the firm within four months following the intervention. We find that interventions significantly reduces worker turnover. Providing managers with worker efficiency information reduces the probability of leaving the firm by 4.5 percentage points, relative to a control mean of 9.7%, corresponding to a reduction of approximately 47 percent. Providing both efficiency and preference information produces a similar effect, reducing turnover by 3.3 percentage points, or approximately 34 percent relative to the control mean. Both estimates are statistically significant at conventional levels, and we cannot reject their equality, with a p-value of 0.396. Thus, we still find no evidence that providing additional preference information generates an incremental reduction in turnover beyond the efficiency information provided in both treatment arms.

[Insert Table 4 Here]

5.3 Results on Knowledge Improvement

The improvement in team efficiency and worker retention suggest that the intervention may have changed how managers make task-assignment decisions. We therefore turn to managers' knowledge of workers' skills and preferences to examine whether the intervention improved managers' ability to identify workers' skills and preferences, thereby enabling them to make more informed and effective task-assignment decisions.

We first examine whether the information intervention improved team managers'

knowledge of workers' skills and preferences. We conducted an in-person knowledge test with all team managers one month after the intervention. The test consisted fo 12 questions, with managers asked to identify workers who matched specific characteristics. For each question, managers were asked to provide three names. The first eight questions assessed knowledge of worker efficiency, including efficiency and learning speed across different levels of task complexity, as well as efficiency variability and performance on teamwork versus solo tasks. The remaining four questions assessed knowledge of worker preferences over task assignments. Appendix Table A1.6 provides a detailed list of these quesions. The ground truth for the efficiency questions was constructed from administrative production records, while the ground truth for the preference questions was based on workers' self-reported preferences from the worker survey.

To construct the post-intervention knowledge measures, a manager receives one point for each correctly identified worker among the top three workers in the corresponding team-by-task cell. We then standardize each question-level score using the control group mean and standard deviation, and construct each outcome as the mean of the standardized scores across its component questions.[7] Thus, the coefficients in Table 5 are expressed in control-group standard deviation units, with a control group mean of zero by construction. The dependent variable in Table 5, Column (1) is the average z-score across the first six questions on worker efficiency and learning speed (Q1–Q6 in Appendix Table A1.6); Column (2) is the average z-score across the two questions on efficiency variability and teamwork (Q7–Q8); and Column (3) is the average z-score across all eight questions related to worker efficiency (Q1–Q8). Column (4) measures knowledge of worker task preferences and is the average z-score across Questions 9–12, while Column (5) is the average z-score across all questions in the knowledge test.

[7]We follow the standardization approach used in Bloom and Van Reenen (2007). Although each question is scored on the same 1-3 scale, the distribution of responses may differ across questions because some management practices may be more difficult to assess than others. Standardizing each practice to have mean zero and standard deviation one before averaging ensures that each question receives equal weight in the composite index, rather than allowing questions with greater variation in their raw responses to disproportionately influence the index.

Table 5, Columns (1)–(3) show that the intervention substantially improved managers' knowledge of worker efficiency. Efficiency information treatment increases the knowledge score for efficiency and learning speed by 0.72 standard deviations, while both efficiency and preference information treatment increases it by 0.81 standard deviations. For the two questions on efficiency variability and teamwork, the both efficiency and preference information treatment is 1.03 standard deviations, while the efficiency information treatment effect is 0.72 standard deviations and is less precisely estimated. The effects on the broader measure combining all eight efficiency-related questions are similarly large, at 0.72 and 0.86 standard deviations for efficiency information treatment and efficiency + preference information treatment, respectively. We cannot reject equality of the two treatment effects for any of these efficiency-related measures, with p-values ranging from 0.554 to 0.787. Thus, both information treatments substantially improved managers' knowledge of worker efficiency, with no statistically detectable difference between the two treatments.

Columns (4) and (5) reveal a different pattern for knowledge of worker preferences and overall knowledge. The efficiency information treatment has a small and statistically insignificant effect on managers' knowledge of worker preferences, increasing the score by only 0.18 standard deviations. In contrast, the efficiency and preference information treatment increases preference knowledge by 1.34 standard deviations, and we strongly reject equality of the two treatment effects ($p = 0.007$). For overall knowledge, the efficiency information treatment increases the score by 0.54 standard deviations, compared with a 1.02-standard-deviation increase under the efficiency and preference information treatment. The difference between the two treatment effects is statistically significant at the 10% level ($p = 0.058$), suggesting that the additional preference information provided under the efficiency and preference information treatment arm generated larger overall gains in managers' knowledge.[8]

[8]Figure 3 compares the distribution of post-intervention knowledge test scores across treatment arms. Control managers cluster tightly around zero, consistent with the standardized index being centered on the control group. Both treatment arms shift the distribution upward. The efficiency information treatment arm has a mean of around +0.6 SD with a fairly wide spread, ranging from about -0.4 to +2.1. The efficiency and preference information treatment arm has a slightly higher mean, near +1 SD, ranging from +0.2 to +1.7, reflecting the additional knowledge gains from preference information.

[Insert Table 5 Here]

We further examine whether the improvement in managers' knowledge varies with their baseline management practices. We measure baseline management practices using 12 survey questions, which are reported in Appendix Table A1.3. We construct a management-practice score from managers' responses and standardize it using the mean and standard deviation of the control group, with higher values indicating better baseline management practices. Table 6 reports heterogeneity in knowledge gains by baseline management practices. Given the relatively small number of teams, we pool the two treatment arms and interact the treatment indicator with an indicator for above-median baseline management practices. The treatment effects are positive across all five knowledge measures, ranging from 0.91 to 1.06 standard deviations for managers with below-median baseline management practices. The interaction terms are negative across all five outcomes, ranging from -0.48 to -0.95, although only the interaction for overall knowledge is statistically significant at the 10% level. This pattern suggests that the intervention may generate larger knowledge gains among managers with weaker baseline management practices.

[Insert Table 6 Here]

5.4 Results on Task Assignment

Having established that the intervention improved managers' knowledge of workers, we next examine whether these knowledge gains translate into changes in task-allocation decisions. We construct a match score to measure the quality of managers' task assignments. Specifically, for each team-day, workers are assigned to task-category cells, where the relevant cell is defined by machine type, complexity level, garment part, or a machine-complexity combination, depending on the outcome family. Each worker-task match is assigned a percentile rank based on the worker's performance relative to other workers who have performed the same task, with higher percentiles

indicating better performance. For each cell, we define the realized match score as the average performance percentile of the workers assigned to that cell. A higher value therefore indicates a better match between workers and tasks.

The realized match score, however, reflects not only the quality of the manager's assignment decision but also the number of positions that must be filled and the pool of workers available to perform the task. Both factors may constrain the quality of feasible assignments independently of managerial decision-making. For example, a team may achieve a high realized match score because relatively few positions need to be filled and many qualified workers are available, even if the manager's assignment decision is no better than usual. Conversely, a team may achieve a low score when demand is high relative to the available pool of qualified workers, forcing the manager to assign less-efficient workers to the task. To account for these constraints and reduce noise in the outcome measure, we benchmark each team-day's realized assignment against the best assignment feasible given that day's task demand and the team's available pool of qualified workers. Specifically, for each cell, we calculate the highest average performance percentile that could be attained given the number of positions to be filled and the team's roster of qualified workers. We then define the cell gap as the difference between the realized match score and this best-feasible benchmark. The gap is non-positive by construction and measures the shortfall of the realized assignment from the best assignment feasible given the day's demand and worker pool. We aggregate these cell-level gaps to the team-day level using worker-hours as weights, yielding a demand-adjusted match score. This measure isolates the component of assignment quality attributable to managers' allocation decisions, conditional on the constraints they face. We describe the detailed construction of the demand-adjusted match score in Appendix A.3.

Table 7 reports treatment effects on the team-day demand-adjusted match score for each of the seven outcome families. Both treatment arms improve task assignment along the efficiency dimension. The efficiency information treatment increases the efficiency match score by 0.041, while the efficiency and preference information treatment increases

it by 0.039; both effects are statistically significant at the 5 percent level. Relative to the control-group mean of -0.298, these estimates correspond to reductions in the average shortfall from the best-feasible benchmark of approximately 13–14 percent. We cannot reject equality of the two treatment effects ($p = 0.94$). Effects on the remaining outcome families are smaller and generally not statistically distinguishable from zero.

We find no significant improvement in assignment quality along the preference dimension even when managers are explicitly provided with information on workers' preferences. The treatment providing both efficiency and preference information raises match scores for machine, complexity-level, and garment-part preferences by 0.044, 0.009, and 0.035, respectively, but none of these effects is statistically significant at conventional levels. The limited use of preference information is consistent with managers' stated priorities and their perceptions of the costs of incorporating worker preferences into task assignment. At baseline, roughly one-quarter of workers reported that their team manager never considered their preferences when assigning tasks, and a similar share were dissatisfied with the tasks assigned to them over the previous three months. At the same time, when asked which factors they consider when assigning tasks, all managers reported considering efficiency-related factors, whereas only 42 percent reported considering worker preferences. Among the 57 percent who reported not considering preferences, approximately 83 percent cited concerns that doing so could reduce team efficiency or make task assignment more difficult. These responses suggest that, when provided with both efficiency and preference information, managers may prioritize the information they view as most important for meeting production objectives. Although incorporating worker preferences could potentially improve workers' satisfaction with their assignments, doing so alongside efficiency considerations may make an already demanding assignment problem more complex. The absence of a detectable effect on preference-based assignment is consistent with managers prioritizing efficiency when faced with multiple dimensions of worker information.

[Insert Table 7 Here]

We further examine heterogeneity in the improvement in task assignment quality along the efficiency dimension. Given the relatively small number of teams, we pool the two treatment arms and interact the treatment indicator with an indicator for below-median baseline knowledge of workers' relative efficiency across tasks. Table 8 reports the results. The effect of the intervention is concentrated among managers with below-median baseline knowledge. For managers with baseline knowledge at or above the median, the estimated treatment effect is small and statistically insignificant (0.015). By contrast, coefficient of the interaction term is 0.067 and statistically significant at the 5% level, indicating a larger treatment effect among managers with less prior knowledge of worker efficiency. This pattern is consistent with information provision generating greater improvements in task assignment among managers who were initially less informed about workers' relative efficiency.

[Insert Table 8 Here]

Taken together, these results suggest that managers selectively incorporate the information provided by the intervention into their task-allocation decisions. Treated managers use information on workers' relative efficiency when assigning workers to tasks, resulting in significant improvements in assignment quality along the efficiency dimension, and these improvements are larger among managers with less prior knowledge of workers' relative efficiency. At the same time, despite substantial improvements in managers' knowledge of other worker characteristics—including learning speed across task types, performance variability across complexity levels, relative performance on multi-worker versus solo tasks, and preferences over different tasks—we find no corresponding improvements in task assignment along these dimensions. This pattern together with evidence from survey data suggests that managers prioritize information they view as particularly important for task assignment, rather than incorporating all available dimensions of worker information. Managers must assign 20–40 or more workers across 30–100 or more tasks within a short period of time, incorpo-

rating multiple dimensions of worker characteristics may make an already complex assignment problem more difficult. More broadly, the results highlight that improving managers' information does not necessarily translate into broader changes in managerial decisions: managers may selectively apply newly acquired information based on the importance they attach to different dimensions of the assignment problem.

5.5 Results on Worker Satisfaction and Happiness

The evidence above shows that the intervention changed managers' task-allocation decisions, particularly by helping them make greater use of information about worker efficiency. We next examine whether these changes in managerial decisions translate into workers' experiences on the job. Using the follow-up worker survey, we study whether the intervention affects overall job satisfaction, satisfaction with task allocation, and workers' evaluations of their team managers.

Table 9 reports the effects of the intervention on three worker-level outcomes constructed from the follow-up survey. Job satisfaction is an index that averages standardized responses across 14 items covering workers' satisfaction with different aspects of their jobs, including workers' satisfaction towards task assignment, their team managers, and the company, as we described in Section 4. Task-allocation satisfaction is the individual item measuring workers' satisfaction with how tasks are assigned. Team-manager evaluation is an index averaging workers' evaluations of their manager along nine dimensions (as we described in Section 4): organizing production, managing conflicts, fairness, listening to workers' voice, caring about workers, communicating clearly, providing guidance, maintaining morale, and being trustworthy. For each outcome, follow-up responses are standardized using the control-group mean and standard deviation at follow-up.

The results in Table 9 show little evidence that the intervention affected workers' overall job satisfaction or satisfaction with task allocation. The efficiency information treatment increases the job satisfaction index by 0.003 standard deviations and task-allocation satisfaction by 0.027 standard deviations; neither estimate is statistically

significant. The corresponding effects of the efficiency and preference information treatment are 0.010 and 0.077 standard deviations, respectively, and are also statistically insignificant. We cannot reject equality of the two treatment effects for either outcome, with p-values of 0.897 and 0.537. In contrast, the efficiency and preference information treatment improves workers' overall evaluation of their team manager by 0.136 standard deviations, marginally significant at the 10 percent level, while the efficiency information treatment has a small and statistically insignificant effect of 0.007 standard deviations. We still cannot reject equality of the two treatment effects at conventional levels ($p = 0.130$). Thus, while the intervention generated measurable changes in managers' task-allocation behavior, these changes do not translate into detectable improvements in workers' overall job satisfaction or satisfaction with task allocation. There is, however, some evidence that providing both efficiency and preference information improves workers' perceptions of their managers.

[Insert Table 9 Here]

Appendix Table A1.7 decomposes the satisfaction and manager-evaluation indices into their underlying survey items. The individual job-satisfaction items show no systematic pattern of treatment effects. The efficiency and preference information treatment increases reported satisfaction with fairness by 0.151 standard deviations. The remaining job-satisfaction items do not show statistically significant differences from zero for either treatment arm. Given that these estimates represent a decomposition of the overall satisfaction index across multiple items, we do not interpret the isolated significant coefficients as evidence of broad changes in workers' job satisfaction.

The individual manager-evaluation items provide somewhat more consistent evidence for the positive effect of the efficiency and preference information treatment. Workers in this treatment arm report higher evaluations of their managers on listening to workers' voices, caring about workers, communicating clearly, and trustworthiness, with effects ranging from 0.140 to 0.182 standard deviations and significant at the 10

percent level. The effects on organizing production, conflict management, fairness, guidance, and morale are positive but not statistically significant. The efficiency information treatment produces small and statistically insignificant effects across all nine manager-evaluation dimensions. Overall, the item-level results are consistent with the composite index: the main detectable improvement is in workers' evaluations of managers under the efficiency and preference information treatment, rather than in workers' overall job satisfaction or satisfaction with task allocation.

5.5.1 Attrition and Sensitivity Analysis

Although we find no statistically significant treatment effects on worker satisfaction or team-manager evaluations, differential attrition across treatment arms raises a potential concern for these outcomes. Appendix Table A1.8 shows that the follow-up rates are 72.6 percent and 75.6 percent in the two treatment arms respectively, compared with 68.8 percent in the control group. Panel B shows that differences in turnover account for a substantial share of this gap. Overall, 12.5 percent of baseline respondents left the firm before the follow-up survey, compared with 16.6 percent in the control group and approximately 10.3 percent in each treatment arm. By contrast, the shares of workers who remained on the roster but did not respond to the follow-up survey are similar across treatment arms, as are the shares whose employment status cannot be determined because their personal information could not be matched to administrative records.

We formally examine the sources of differential follow-up in Appendix Table A1.9. Both treatment arms are more likely to be re-surveyed than the control group, although the difference is statistically significant only for the efficiency and preference information treatment. The treatment effects on the probability of leaving the firm in the baseline respondents sample are -0.0649 and -0.0722 for the two treatment arms, respectively, both statistically significant. In contrast, we find no statistically significant treatment effects on non-response among workers who remained employed or on unmatched workers. These results indicate that the higher follow-up rates in the treatment arms are largely attributable to the intervention's reduction in worker turnover, rather than differences in survey response among workers who remained employed. Consequently,

the composition of workers observed at follow-up may differ across treatment arms: the intervention may retain workers who would otherwise have left the firm.

This differential retention could affect our estimates of worker satisfaction if workers who would otherwise leave differ systematically from those who remain. To assess this possibility, Appendix Table A1.10 compares baseline satisfaction and team-manager evaluations between workers who remained with the firm and those who left before the follow-up survey. Leavers report significantly lower satisfaction with promotion opportunities, higher quit intentions, lower wage satisfaction, and a lower willingness to remain with the firm in the long term. They also have lower point estimates for task-allocation satisfaction and the overall job satisfaction index, although these differences are not statistically significant. These patterns suggest that workers retained by the intervention may be negatively selected in terms of baseline satisfaction, which could attenuate estimated treatment effects on satisfaction at follow-up. The direction and magnitude of this selection, however, cannot be established conclusively from these comparisons alone.

We therefore conduct a Lee (2009) bounds analysis to assess the sensitivity of the satisfaction results to differential attrition. Appendix Table A1.11 reports bounds separately for each treatment arm relative to the control group, trimming the treated outcome distribution by the difference in follow-up rates under the monotonicity assumption that treatment affects selection in only one direction. Although the resulting bounds include zero for job satisfaction in both treatment arms and for task-allocation satisfaction and team-manager evaluations in their lower bounds, the upper bounds permit moderately positive effects on task-allocation satisfaction and team-manager evaluations. Thus, while differential attrition introduces uncertainty into the interpretation of the null estimates, the bounds do not rule out modest positive effects on some worker job satisfaction and team-manager evaluation outcomes.

6 Conclusion

We use a randomized field experiment in three large garment manufacturing plants in China to examine whether reducing information frictions can improve managers' task-allocation decisions. We find that providing managers with information about workers' efficiency substantially improves their knowledge of worker skills and leads them to make better assignments along the efficiency dimension. These improvements translate into economically meaningful gains in team efficiency and worker retention, without reducing product quality. Providing additional information about worker preferences substantially improves managers' knowledge of those preferences, but does not generate corresponding improvements in preference-based task assignment. Consistent with this selective use of information, we find no detectable effects on workers' overall job satisfaction or satisfaction with task allocation, although the additional preference information treatment generates a marginal improvement in workers' evaluations of their managers. Taken together, the results show that reducing information frictions can improve managerial decision-making and organizational performance, but that managers do not necessarily incorporate all dimensions of available information into complex allocation decisions.

These findings raise an important question for future research: can algorithm-based decision-support systems overcome some of the information-processing constraints faced by human managers in complex task-allocation environments? Managers in our setting do not simultaneously consider multiple dimensions of worker characteristics while assigning 20-40 or more workers across many tasks under tight time constraints. The selective use of information that we observe suggests that the challenge may not be simply access to information, but the capacity to process and apply it in real time. Algorithms may be able to integrate more dimensions of worker information and generate recommendations at a scale that is difficult for individual managers to match. However, whether algorithm can outperform human managers is an empirical question. Managers may possess tacit knowledge, contextual information, or observations about workers that are not captured in administrative data, and they may also be reluctant to

follow algorithmic recommendations when those recommendations conflict with their own judgment. Understanding when algorithms complements managerial knowledge, when it substitutes for it, how managers respond to algorithmic recommendations, and how this can improve management of firms in developing context and therefore translates into improved worker welfare is therefore an important direction for future research.

In terms of the policy implications of the study, our findings suggest that managers in manufacturing plants in developing countries often face substantial constraints in making complex task-assignment decisions within a short period of time, and that providing managers with better information can be an effective way to facilitate task assignment and improve management practices. At the same time, our results reveal that worker preferences are often not an important consideration in managers' task-assignment decisions, which may have implications for workers' experiences on the job. Simply making preference information available may not be sufficient to change managerial practices when managers perceive a trade-off between accommodating worker preferences and maintaining team efficiency. Finally, our experience reaching out to dozens of other similar large-scale garment manufacturing plants suggests that several foundational conditions may need to be in place before firms can effectively implement such information-based interventions. Some plants do not maintain the administrative data necessary to analyze worker skills and preferences; others have managers who do not perceive the value of this type of managerial assistance; and some experience sufficiently high worker turnover and absenteeism that collecting the longer-term data needed to characterize workers is difficult. Taken together, these observations suggest that while information provision can be effective in improving management practices, successful implementation at scale may also require strengthening data infrastructure and providing both managerial and worker training to enable firms to adopt and sustain better management practices in developing-country contexts.

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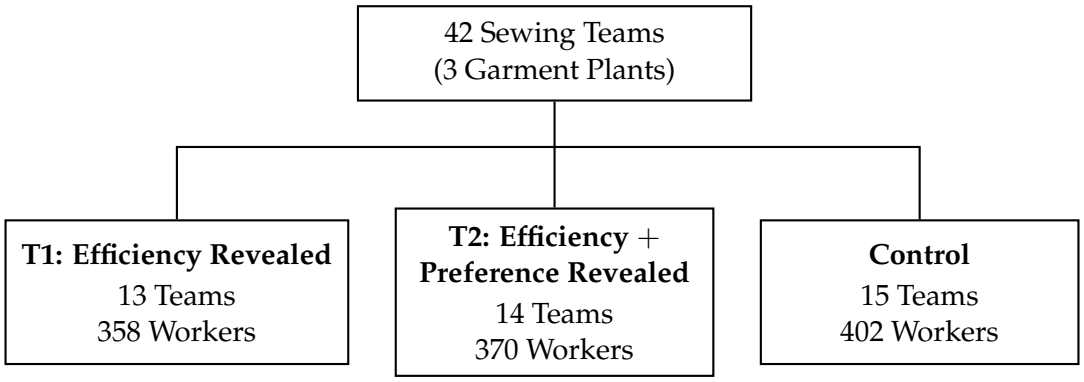
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Figure 1: Randomization Design



Note: The randomization is conducted at the team level and stratified by team size and performance within each plant.

Figure 2: Intervention Timeline

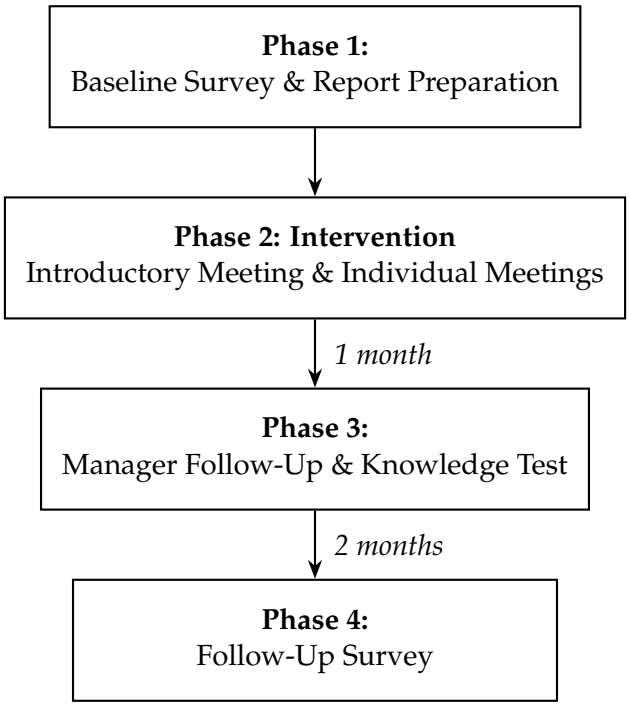


Figure 3: Distribution of Post-Intervention Knowledge Test Scores

This figure displays the distribution of knowledge-test z-scores constructed from the follow-up knowledge test across treatment arms.

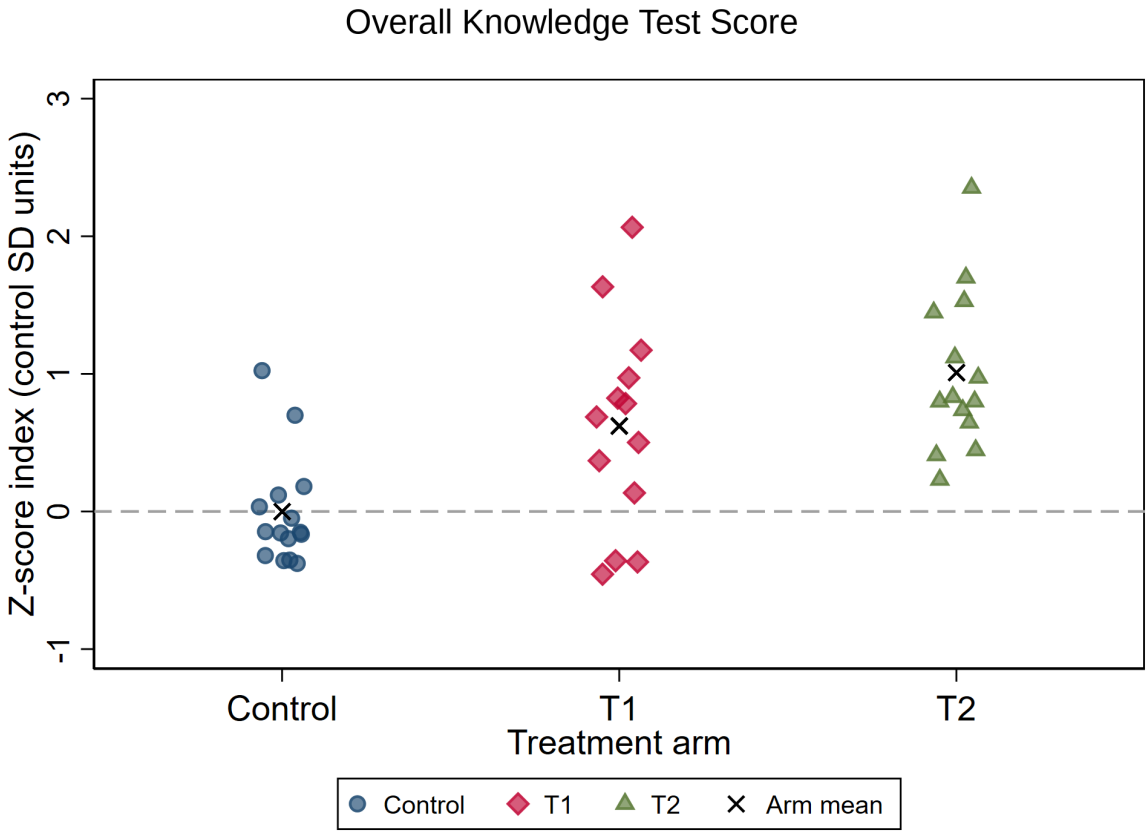


Table 1: Pre-intervention Summary Statistics for the Administrative and Survey Data

	(1) Control	(2) Treatment 1	(3) Treatment 2	(4) Difference (2)-(1)	(5) Difference (3)-(1)	(6) N
Manager characteristics						
Age	44.667	41.692	41.071	-2.974	-3.595	42
	(8.200)	(4.679)	(5.811)	(0.238)	(0.179)	
Female	0.667	0.692	0.571	0.026	-0.095	42
	(0.488)	(0.480)	(0.514)	(0.889)	(0.612)	
Tenure	5.023	3.980	4.038	-1.043	-0.985	42
	(2.251)	(2.463)	(2.660)	(0.251)	(0.290)	
Education(yrs)	9.400	9.462	10.071	0.062	0.671	42
	(1.549)	(1.127)	(1.900)	(0.904)	(0.305)	
Team composition						
Age	43.370	43.038	42.858	-0.331	-0.512	42
	(2.395)	(2.234)	(2.443)	(0.707)	(0.573)	
Female	0.873	0.912	0.904	0.039	0.031	42
	(0.083)	(0.052)	(0.064)	(0.135)	(0.270)	
Tenure	3.184	3.409	3.331	0.226	0.148	42
	(1.203)	(1.160)	(1.184)	(0.617)	(0.741)	
Education(yrs)	9.130	9.251	9.217	0.120	0.086	42
	(0.250)	(0.359)	(0.244)	(0.317)	(0.353)	
Team Performance						
Size	29.467	30.462	28.143	0.995	-1.324	42
	(8.184)	(8.038)	(6.826)	(0.748)	(0.638)	
TurnOver	0.162	0.156	0.137	-0.006	-0.025	42
	(0.111)	(0.071)	(0.087)	(0.873)	(0.510)	
Productivity	0.871	0.860	0.868	-0.011	-0.003	42
	(0.171)	(0.169)	(0.183)	(0.863)	(0.967)	
Quality	0.862	0.863	0.865	0.001	0.003	42
	(0.027)	(0.060)	(0.032)	(0.954)	(0.799)	
Bonus	945.035	898.600	919.680	-46.435	-25.355	42
	(806.030)	(702.930)	(776.799)	(0.871)	(0.932)	
Survey Data						
Satisfaction	-0.000	-0.032	-0.018	-0.032	-0.018	1086
	(0.515)	(0.552)	(0.508)	(0.562)	(0.724)	
Evaluation	-0.000	-0.089	0.034	-0.089	0.034	1086
	(0.862)	(0.943)	(0.835)	(0.442)	(0.639)	

Notes: Columns (1) to (3) report means for each group, with standard deviations in parentheses. Columns (4) and (5) present coefficients from regressions of each outcome variable on treatment indicators, with p-values in parentheses. The control group serves as the baseline category in all regressions. Satisfaction and Evaluation are composite z-scores: each of the 15 satisfaction and 9 team-manager evaluation survey items is standardized by subtracting the control-group mean and dividing by the control-group standard deviation, and the item z-scores are then averaged.

Table 2: Summary Statistics: Managers' Baseline Knowledge-Test Scores

	(1) Observations	(2) Mean	(3) Min	(4) Max	(5) Median
Efficiency Knowledge	42	2.714	0	4	3
Variability Knowledge	42	0.833	0	3	1

Notes: Each row is a team-level score, from the managers' baseline knowledge test: the number of the manager's named workers who are actually in the team's ground-truth top-5. For the ground-truth of efficiency knowledge, we first use efficiency measure by machine and complexity level calculated using the past one year output data. We first take the flatbed sewing machine-A workers in ranking order; if fewer than five names have been accumulated, we continue with the flatbed sewing machine-B workers in their own ranking order, skipping any workers already taken from flatbed sewing machine-A. Since level A is the most complex task, we treat the workers ranked in the top five on the most complex task as the most efficient workers in that machine category. To obtain the ground truth for variability knowledge, we first calculate the downside volatility of each worker-task pair, then pooled into a single score per worker, weighted by the number of observations for each task. We rank workers within each team by overall variability, with rank 1 denote the most volatile worker. The top-5 ground truth consists of workers with a within-team rank of 5 or lower.

Table 3: Effect of Information on Team Efficiency and Team Quality

	(1) Raw Efficiency	(2) Complexity-Adjusted Efficiency	(3) Quality
Treatment 1	0.0980**	0.0848**	-0.0074
	(0.0439)	(0.0376)	(0.0239)
Treatment 2	0.0616	0.0806**	0.0014
	(0.0419)	(0.0354)	(0.0156)
Control mean	0.5887	0.7915	0.8346
p-value (T1 = T2)	0.373	0.906	0.725
Observations	450	450	116
Clusters (teams)	42	42	42
Randomization batch x stratum FE	Yes	Yes	Yes
Time fixed effects	Plant x week	Plant x week	Plant x month
Equal team weights	Yes	Yes	Yes
Controls for pre-intervention mean	Yes	Yes	Yes

Notes: The sample is post-intervention team-weeks, event weeks 0-10. Dependent variable in Column (1) is the raw weekly efficiency, and dependent variable in Column (2) is raw weekly efficiency net of style complexity, where complexity is the number of distinct operations of the style performed by the team, and its coefficients are estimated on pre-intervention team-days only then held fixed. Log style tenure is included in that first-stage regression, so the complexity coefficients are not confounded by learning, but it is not subtracted, any treatment effect operating through the learning curve therefore remains in the outcome. We include plant by calendar-week fixed effects and randomization batch x stratum fixed effects throughout, and weight by the inverse of the team's number of observed post-intervention weeks, so each team contributes equal total weight. We control pre-intervention mean in all regressions, which is the team's average over event weeks-11 to-1, an 11-week baseline window matching the 11-week post window. Standard errors clustered by team in parentheses; stars use those cluster-robust standard errors. *** p<0.01, ** p<0.05, * p<0.10.

Table 4: Effect of Information on Worker Turnover

	(1) Workers Leaving Firms
Treatment 1	-0.0453***
	(0.0147)
Treatment 2	-0.0333**
	(0.0128)
Control Mean	0.0970
p-value (T1=T2)	0.396
Number of Clusters	42
Observations	1130

Note: The sample includes workers in roster who had not already left at the time of intervention. The dependent variable is an indicator for whether the worker left the company within 4 months after the intervention. We control for randomization batch by strata fixed effects, and the team's turnover rate in the 3-month window prior to the intervention. Standard errors clustered by team in parentheses. *** p<0.01, ** p<0.05, * p<0.10.

Table 5: Effect of Information Treatment on Manager Knowledge

	(1) Efficiency	(2) Variability & Teamwork	(3) All Efficiency Questions	(4) Preference	(5) Overall Knowledge Test Scores
T1 (Efficiency treatment)	0.7184**	0.7165	0.7180**	0.1816	0.5392**
	(0.3149)	(0.4457)	(0.2980)	(0.3469)	(0.2468)
T2 (Efficiency + preference treatment)	0.8051***	1.0288**	0.8610***	1.3371***	1.0197***
	(0.2019)	(0.4076)	(0.1901)	(0.3560)	(0.1955)
Control Mean	0.0000	0.0000	0.0000	0.0000	0.0000
Test T1=T2 (p-value)	0.787	0.554	0.627	0.007	0.058
Number of Clusters	42	42	42	42	42
Observations	42	42	42	42	42

Notes. Columns (1)-(5) report the average z-scores for efficiency and learning speed, efficiency variability and teamwork, all efficiency-related questions, worker task preferences, and overall knowledge, respectively. Each question-level score is standardized using the control-group mean and standard deviation, and each dependent variable is the mean of the standardized scores across its component questions. Thus, coefficients are expressed in control-group standard deviation units, and the control-group mean is 0 by construction. For scoring rule, managers receive one point for each correctly identified worker in the corresponding team-by-task cell. All columns include randomization batch $\times$ stratum fixed effects. Standard errors clustered by team are reported in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Heterogeneous Effects of Treatment on Manager Knowledge by Management Practice Scores (Pool Treatment)

	(1) Efficiency	(2) Variability & Teamwork	(3) All Efficiency Questions	(4) Preference	(5) Overall Knowledge Test Scores
Treatment	1.001***	0.908**	0.977***	1.056**	1.004***
	(0.370)	(0.367)	(0.324)	(0.462)	(0.255)
High baseline mgmt practice z-score (>plant median)	0.186	1.068**	0.406*	1.233**	0.682***
	(0.276)	(0.430)	(0.227)	(0.502)	(0.250)
Treated x High baseline mgmt practice z-score (>plant median)	-0.481	-0.509	-0.488	-0.950	-0.642*
	(0.427)	(0.635)	(0.369)	(0.680)	(0.334)
Control mean (>plant median)	0.105	0.459	0.194	0.602	0.330
Observations	42	42	42	42	42

Note: This table examines whether the improvement in managers' knowledge differs by baseline management practices. The dependent variable is the z-score of managers' knowledge measured in the follow-up knowledge test, constructed as in Table 5. Treatment is an indicator equal to 1 for teams assigned to either treatment arm and 0 for control teams. High baseline management practice is an indicator equal to 1 if the team manager's baseline management-practice z-score is above the plant-specific median and 0 otherwise. All specifications include randomization batch $\times$ stratum fixed effects. Standard errors clustered at the team level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Effects on Task Assignment

	Efficiency	Learning	Variability	Multi-worker	Preference	Machine	Level	Part
Treatment 1	0.0408**	0.0248	-0.0348	-0.0146	0.0112	-0.0017	-0.0342	0.0094
	(0.0192)	(0.0232)	(0.0211)	(0.0208)	(0.0210)	(0.0296)	(0.0220)	(0.0325)
Treatment 2	0.0375**	0.0214	-0.0111	0.0042	0.0366	0.0443	0.0086	0.0351
	(0.0156)	(0.0195)	(0.0191)	(0.0233)	(0.0238)	(0.0304)	(0.0291)	(0.0365)
Control mean	-0.2981	-0.3016	-0.2816	-0.2655	-0.3924	-0.1428	-0.3783	-0.4321
p-value: T1=T2	0.8792	0.8847	0.2888	0.3884	0.2993	0.1035	0.0628	0.4789
Observations	2522	2523	2523	2398	2521	2520	2519	2519
Clusters (teams)	42	42	42	40	42	42	42	42
Randomisation batch $\times$ stratum FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Plant $\times$ week FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Equal team weights	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls for pre-intervention mean	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable: team-day demand-adjusted match score for each outcome family, defined as the team's actual match score minus the best-feasible benchmark match score attainable given that day's task mix; higher values indicate task assignments closer to the demand-adjusted benchmark. Control mean reports the raw, unweighted mean of the dependent variable among Control-arm team-days during the post-intervention period. All regressions include randomization-strata and plant $\times$ calendar-week fixed effects, control for the team's pre-intervention mean of the outcome, and weight by the inverse number of team-days so that every team contributes equally. Clustered-robust standard errors by team are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Efficiency Adoption Heterogeneity by Below-Median Baseline Knowledge (Pooled Treatment)

	Match Score (Efficiency)
Treatment	0.0148
	(0.0175)
Treatment x Low Score	0.0674**
	(0.0308)
Low Score	-0.0354
	(0.0250)
Observations	2522
Clusters (teams)	42
Plant x week FE	Yes
Randomization-Stratum FE	Yes
Equal team weights	Yes

Notes: The dependent variable is the demand-adjusted match score, as in Column (1) of Table 7. Treatment is an indicator equal to one for teams assigned to either treatment arm and zero for control teams. Low score is an indicator equal to one if the team manager's baseline knowledge score on the efficiency dimension is below the sample median (median = 3), and zero otherwise. The Treatment coefficient captures the treatment effect for managers with baseline knowledge at or above the median, while the interaction coefficient captures the additional treatment effect for managers with below-median baseline knowledge. All regressions control for randomization-stratum and plant $\times$ calendar-week fixed effects and use equal weights across teams. Cluster-robust standard errors by team are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Effect of Information on Worker Survey Outcomes

	(1) Job satisfaction	(2) Task-allocation sat.	(3) Team-manager eval.
Treatment 1	0.0029	0.0269	0.0074
	(0.0461)	(0.0885)	(0.0841)
Treatment 2	0.0100	0.0771	0.1358*
	(0.0501)	(0.0899)	(0.0779)
Control mean	-0.0000	-0.0000	0.0000
p-value (T1=T2)	0.897	0.537	0.130
Observations	823	823	823
Clusters (teams)	42	42	42

Note: One observation per worker re-surveyed at follow-up. Outcomes are z-scored using the mean and standard deviation of the control group at the follow-up survey; the baseline z-score control is separately z-scored using the mean and standard deviation of the control group at baseline. Every column controls for the worker's baseline z-score of the same outcome. Randomisation batch x stratum fixed effects are included. Standard errors clustered by team in parentheses; stars are based on those cluster-robust standard errors. Because treated workers were more likely to be re-surveyed, these estimate the effect among workers observed at follow-up; see the accompanying Lee bounds. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

Appendix

Enhancing Decision-Making: The Impact of Information Provision on Worker Task Assignment

Yunzhu Lai
Agricultural and Resource Economics, University of Maryland

A.1 Additional Tables and Figures

Table A1.1: Job Satisfaction Survey Items

No.	Statement
1	Overall, I am satisfied with the tasks assigned to me in the past three months.
2	My supervisor does not consider the feelings of subordinates.
3	I have not been treated fairly by my supervisor.
4	If I feel that my task allocation is unfair, I will calmly communicate with my supervisor.
5	I want to transfer to another sewing team/production line.
6	I feel that my efforts have not been adequately rewarded.
7	My workload is too heavy.
8	My promotion opportunities are too limited.
9	Due to the limited abilities of my colleagues, my workload has increased.
10	I sometimes feel that my work is meaningless.
11	I am frequently criticized by my supervisor.
12	In the past three months, I have considered resigning from the company multiple times.
13	Overall, I am satisfied with my salary income over the past three months.
14	I am willing to work at this company for the long term (e.g., 3 years or more).

Note: Respondents indicated their agreement with each statement on a 5-point scale ranging from Strongly Disagree (1) to Strongly Agree (5).

Table A1.2: Worker Evaluation of Team Manager Performance

No.	Item
1	Organizing production
2	Handling emergencies and conflicts
3	Fairness
4	Listening to workers' opinions and suggestions
5	Caring about workers' work and life
6	Clearly communicates the requirements of my job
7	Provides continued guidance on my work
8	Inspires team morale when facing challenges
9	Trustworthey

Note: Respondents rated their team manager's performance over the past three months on a 5-point scale ranging from Very Poor (1) to Very Good (5).

Table A1.3: Manager Survey: Management Practices

No.	Question
Retrospective Analysis and Employee Management	
1	How often do you do the following retrospective analysis: What went right and how can I assure it will happen again?
2	How often do you do the following retrospective analysis: What are the issues I can avoid being repeated next time?
3	How often do you do the following retrospective analysis: What can I do to ensure employees see me as the best manager possible?
4	How often do you do the following retrospective analysis: What can I do to motivate my employees better?
5	If you find that an employee is performing poorly (possibly due to a lack of motivation or insufficient skills), what measures would you take to address the issue?
6	If a high-performing employee plans to resign, what measures would you take to persuade them to stay?
Management Philosophy	
12.1	Maintaining daily communication with workers helps identify problems and improve efficiency.
12.2	As long as there are no obvious problems, I usually do not need to adjust my task allocation strategy.
12.3	I tend to make management decisions based on past experience.
12.4	As long as workers complete tasks as required, they do not need to express their opinions.
12.5	I am used to communicating with subordinate workers in an authoritative, unquestionable tone.
12.6	In the past three months, I have proactively inquired about situations where workers' physical health, emotions, or family matters affected their work.

Note: Items 1-4 use a 6-point frequency scale (Never, Less than once a month, Once a month, Once a week, Every other day, Every day); items 5-6 are check-all-that-apply; items 12.1-12.6 use a 5-point agreement scale ranging from Strongly Disagree (1) to Strongly Agree (5).

Table A1.4: Garment Complexity and Raw Daily Team Efficiency, Pre-Intervention Sample

	(1) Raw Team Efficiency
Number of operations	-0.002327**
	(0.000895)
Log style tenure	0.131345***
	(0.010403)
Plant fixed effects	Yes
Observations	14,168
Clusters (teams)	42
Adjusted R-squared	0.3853

Notes: OLS regression of raw daily team efficiency on the number of distinct tasks in the team's style that day, a garment-level complexity measure, and log style tenure, estimated on each plant's pre-intervention team-days only. Clustered standard errors by team in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

Table A1.5: Effect of Information on Team Efficiency: Style FE vs. Style FE + Tenure

	Style FE Only	Style FE + Log Tenure
Treatment 1	0.0565*	0.0592**
	(0.0280)	(0.0249)
Treatment 2	0.0714**	0.0561**
	(0.0286)	(0.0236)
Control mean	0.6422	0.6321
p-value: T1 = T2	0.5795	0.8893
Observations	450	450
Clusters (teams)	42	42
Randomisation batch x stratum FE	Yes	Yes
Plant x week FE	Yes	Yes
Equal team weights	Yes	Yes
Controls for pre-intervention mean	Yes	Yes

Note: The sample is post-intervention team-weeks, event weeks 0-10. The dependent variables in Column (1) and Columns (2) are weekly team efficiency residualized on style fixed effects and weekly team efficiency residualized on both style fixed effects and log style tenure that accounts for the learning rate of teams. Pre-intervention mean is the team's average of the same measure over its 11 pre-intervention weeks. Randomization batch $\times$ stratum fixed effects and plant $\times$ calendar week fixed effects are included in all columns. We also weight observations by the inverse of the team's number of observed post-intervention weeks, so every team contributes the same total weight. Standard errors clustered by team in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

Table A1.6: Follow-up Knowledge Test Items

No.	Question
Part 1 – Efficiency Questions	
1	List the three workers in your team who are most efficient at operating flatbed sewing A-level, ranked from highest to lowest efficiency.
2	List the three workers in your team who have the highest learning speed when performing flatbed sewing A-level tasks at the start of a new style, ranked from fastest to slowest learning speed.
3	List the three workers in your team who are most efficient at operating flatbed sewing B-level tasks, ranked from highest to lowest efficiency.
4	List the three workers in your team who have the highest learning speed when performing flatbed sewing B-level tasks at the start of a new style, ranked from fastest to slowest learning speed.
5	List the three workers in your team who are most efficient at operating flatbed sewing C-level tasks, ranked from highest to lowest efficiency.
6	List the three workers in your team who have the highest learning speed when performing flatbed sewing C-level tasks at the start of a new style, ranked from fastest to slowest learning speed.
7	List the three workers in your team whose efficiency on B-level tasks fluctuates the most (i.e., frequently and noticeably falls below their personal average level), ranked from highest to lowest degree of fluctuation.
8	List the names of workers in your team whose efficiency is noticeably higher when performing multi-worker tasks than when working alone.
Part 2 – Preference Questions	
9	List the names of three workers in your team who prefer performing flatbed sewing A-level tasks.
10	List the names of three workers in your team who prefer performing flatbed sewing B-level tasks.
11	Suppose there were a staff shortage for A-level tasks — among workers who do not normally do A-level work, who do you think would be willing to fill in? Please give the names of three workers.
12	Please list the names of three workers in your team who prefer doing pocket-related tasks.

Table A1.7: Effect of Information on Individual Survey Items

	Treatment 1	Treatment 2	p(T1=T2)
Panel A: Job satisfaction items			
Task assignment	0.0269	0.0771	0.5375
	(0.0885)	(0.0899)	
Empathy	-0.0006	0.0146	0.8388
	(0.0751)	(0.0752)	
Fairness	0.1012	0.1514*	0.5741
	(0.0938)	(0.0859)	
Voice	0.0394	0.0193	0.7496
	(0.0808)	(0.0770)	
Transfer	0.0668	0.0133	0.5901
	(0.0874)	(0.0996)	
Reward effort	0.0192	-0.1070	0.2252
	(0.0896)	(0.0843)	
Workload	-0.1165	0.0478	0.0397
	(0.0862)	(0.0839)	
Promotion	0.0671	0.0103	0.5154
	(0.0845)	(0.1058)	
Colleagues' ability	-0.0239	-0.0413	0.7851
	(0.0628)	(0.0710)	
Job meaning	-0.1434	-0.0432	0.3108
	(0.0878)	(0.0880)	
Criticism	0.0719	-0.0214	0.3261
	(0.0909)	(0.0860)	
Quit	-0.1570	-0.0222	0.1579
	(0.0980)	(0.0713)	
Wage	-0.0087	0.0451	0.7627
	(0.1329)	(0.1480)	
Tenure	0.0760	0.0024	0.4728
	(0.1036)	(0.0796)	
Welfare	-0.0734	-0.0057	0.5778
	(0.1219)	(0.1142)	
Panel B: Team-Manager evaluation items			
Organize production	-0.0189	0.1371	0.1327
	(0.1107)	(0.1026)	
Conflict management	0.0006	0.0662	0.4527
	(0.0871)	(0.0755)	
Fairness	0.0131	0.0650	0.5788
	(0.1014)	(0.0881)	
Listen	-0.0136	0.1408*	0.0550
	(0.0902)	(0.0744)	
Care	0.0382	0.1514*	0.1797
	(0.0853)	(0.0791)	
Communicate clearly	-0.0004	0.1821*	0.0699
	(0.1041)	(0.1017)	
Guidance	0.0186	0.1596	0.2176
	(0.1024)	(0.1016)	
Morale	-0.0253	0.1364	0.1443
	(0.1061)	(0.0953)	
Trustworthy	0.0040	0.1709*	0.1043
	(0.0911)	(0.0887)	

Notes. Each row reports a separate regression of one survey item on the two treatment indicators. All regressions control for the worker's baseline value of the same item and randomization batch $\times$ stratum fixed effects. Follow-up items are standardized using the control-group mean and standard deviation at follow-up; baseline items are standardized using the corresponding control-group distribution at baseline. One observation per worker re-surveyed at follow-up. Standard errors clustered by team are reported in parentheses. The last column reports the cluster-robust p-value for the test of equality between the two treatment coefficients. The 24 regressions decompose the composite indices reported in the main text. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A1.8: Decomposition of Survey Non-Response at Follow-Up

	Control	T1	T2	All
Baseline respondents at risk	404	387	349	1140
Panel A: Follow-up status				
Followed	278	281	264	823
	(68.8%)	(72.6%)	(75.6%)	(72.2%)
Not followed	126	106	85	317
	(31.2%)	(27.4%)	(24.4%)	(27.8%)
Panel B: Decomposition of non-follow-up				
Left the firm	67	40	36	143
	(16.6%)	(10.3%)	(10.3%)	(12.5%)
Still on roster, not re-surveyed	41	35	34	110
	(10.1%)	(9.0%)	(9.7%)	(9.6%)
Left status unknown	18	31	15	64
	(4.5%)	(8.0%)	(4.3%)	(5.6%)

Note: This table decomposes survey non-response at follow-up by treatment arm. The sample includes all respondents to the baseline survey. Panel A reports the proportions of baseline respondents who were successfully followed at follow-up and those whom we were unable to follow. Panel B further decomposes non-follow-up into three categories: respondents who left the company before the follow-up survey; respondents who remained on the company roster but did not respond to the follow-up survey; and respondents who provided incorrect personal information, preventing us from matching them to the company roster and determining their employment status. Percentages in parentheses are calculated relative to the number of baseline respondents in each treatment arm.

Table A1.9: What Accounts for the Differential Follow-Up Rate?

	(1)	(2)	(3)	(4)
	Re-surveyed	Not re-surveyed, because:		
	at follow-up	Left firm	Still employed	Unmatched
Treatment 1	0.0540	-0.0649**	-0.0216	0.0326
	(0.0501)	(0.0292)	(0.0345)	(0.0204)
Treatment 2	0.0853**	-0.0722***	-0.0099	-0.0031
	(0.0418)	(0.0260)	(0.0314)	(0.0157)
Control mean	0.6881	0.1658	0.1015	0.0446
p-value (T1 = T2)	0.453	0.780	0.717	0.103
Observations	1140	1140	1140	1140
Clusters (teams)	42	42	42	42

Note: This table examines the sources of differential follow-up non-response across treatment arms. Each column reports a separate OLS regression of the indicated outcome on treatment-arm indicators, using all baseline survey respondents. The dependent variables are mutually exclusive and exhaustive indicators that partition the baseline sample. The dependent variable of Column (2) indicates that the respondent was not re-surveyed and was recorded as having left the firm before the follow-up survey. The dependent variable of Column (3) indicates that the respondent was not re-surveyed but remained on the firm roster, representing survey non-response among workers who remained employed. The dependent variable of Column (4) indicates that the respondent was not re-surveyed and could not be matched to administrative records due to reporting errors, so employment status cannot be determined. All regressions include randomization batch-by-stratum fixed effects. Standard errors, clustered at the team level, are reported in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

Table A1.10: Baseline Satisfaction and Team-Leader Evaluation of Stayers and Leavers

	Stayers (1)	Leavers (2)	Difference (3)	p-value (4)	Obs. (S / L) (5)
Panel A: Indices					
Job satisfaction index (15 items)	0.019	-0.077	-0.095	0.142	908 / 152
Task-allocation satisfaction	0.057	-0.046	-0.103	0.182	908 / 152
Team-leader evaluation index (9 items)	-0.014	0.037	0.050	0.577	908 / 152
Panel B: Job-satisfaction items					
Task allocation	0.057	-0.046	-0.103	0.182	908 / 152
Supervisor empathy	0.039	0.115	0.076	0.525	908 / 152
Fairness	-0.017	0.044	0.060	0.595	908 / 152
Voice	-0.030	-0.020	0.010	0.924	908 / 152
Transfer	0.020	0.082	0.062	0.594	908 / 152
Effort	0.003	-0.130	-0.133	0.245	908 / 152
Workload	-0.040	-0.168	-0.129	0.208	908 / 152
Promotion	0.029	-0.241	-0.269***	0.008	908 / 152
Colleagues	-0.016	-0.117	-0.102	0.309	908 / 152
Meaningfulness of work	0.030	-0.007	-0.036	0.720	908 / 152
Criticism	0.018	0.053	0.035	0.739	908 / 152
Quit intention	0.105	-0.202	-0.307***	0.009	908 / 152
Wage	0.026	-0.176	-0.202*	0.091	908 / 152
Wants to stay long term (3+ yrs)	0.144	-0.278	-0.422***	0.000	908 / 152
Welfare	-0.089	-0.058	0.031	0.809	908 / 152
Panel C: Team-leader evaluation items					
Organises production	-0.009	0.055	0.064	0.498	908 / 152
Manages conflict	-0.025	-0.002	0.023	0.827	908 / 152
Fairness	-0.002	0.065	0.067	0.475	908 / 152
Listens	0.001	0.068	0.067	0.483	908 / 152
Cares about workers	-0.030	0.028	0.059	0.596	908 / 152
Gives clear instructions	-0.007	0.095	0.102	0.411	908 / 152
Provides guidance	-0.007	0.069	0.076	0.498	908 / 152
Boosts morale	-0.009	-0.022	-0.013	0.896	908 / 152
Trustworthy	-0.034	-0.026	0.008	0.936	908 / 152

Note: Sample is baseline survey respondents matched to the turnover panel (908 stayers, 152 leavers, 42 teams). Leavers are workers who showed up in the baseline survey but left the company before the follow-up survey; stayers did not. All variables are baseline values in z-score units, standardized on the baseline control group; indices are the unweighted mean of item z-scores. Column (3) is the difference between leavers and stayers. Column (4) is the p-value of a regression of the variable on a leaver dummy with standard errors clustered by team. Column (5) reports the number of non-missing stayers (S) and leavers (L). *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

Table A1.11: Lee (2009) Bounds on the Effect of Information on Worker Survey Outcomes

Panel A: Follow-up rate by arm			
	Baseline respondents	Follow-up rate	Trim share
Control	404	0.6881	—
Treatment 1	387	0.7261	0.0523
Treatment 2	349	0.7564	0.0903
Panel B: Trimmed ANCOVA bounds			
	Job satisfaction	Task-allocation sat.	Team-leader eval.
Treatment 1: Lower bound	-0.0685	-0.0418	-0.0566
	(0.0411)	(0.0899)	(0.0784)
Treatment 1: Upper bound	0.0245	0.1771**	0.0795
	(0.0417)	(0.0829)	(0.0760)
Treatment 2: Lower bound	-0.0712*	-0.0424	0.0423
	(0.0378)	(0.0780)	(0.0715)
Treatment 2: Upper bound	0.0749	0.2680**	0.2480***
	(0.0505)	(0.0982)	(0.0758)
Observations (T1 vs control)	544	544	544
Observations (T2 vs control)	518	518	518

Note: Panel A reports the share of baseline respondents in each treatment arm who were successfully re-surveyed at follow-up. Because follow-up rates are higher in the treated arms, we trim the outcome distributions of each treated arm by the fraction shown in the final column, $(s_a - s_c)/s_a$, where s_a denotes the follow-up rate in treated arm a and s_c denotes the follow-up rate in the control arm. The control distribution is left unchanged. Panel B reports Lee bounds obtained after trimming. For each treated arm, we estimate the treatment effect separately against the control group, excluding the other treated arm because Lee bounds are defined for binary treatment comparisons. Hence observations are the estimation sample for each pair of bounds: control and the indicated treated arm only, restricted to workers re-surveyed at follow-up and net of the trimmed observations. Each specification regresses the outcome on a treatment indicator, the baseline z-score of the same outcome, and randomization batch $\times$ stratum fixed effects; standard errors are clustered by team. The lower bound trims the highest outcomes from the treated distribution, corresponding to the assumption that workers retained because of treatment have the highest satisfaction or evaluation scores. The upper bound trims the lowest outcomes, corresponding to the assumption that these workers have the lowest scores. Ties are broken randomly using a fixed seed. Identification requires monotonicity, i.e., that treatment affects selection in only one direction; the estimated reduction in turnover is consistent with this assumption. Statistical significance is based on cluster-robust standard errors for the corresponding bound. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.2 Construction of Team Efficiency Measures

This appendix describes how we construct the three team efficiency measures used in the paper: raw team efficiency, team efficiency residualized on style fixed effects and the learning rate, and complexity-adjusted team efficiency. We also explain why the adjusted measures, rather than raw efficiency, are our preferred outcomes.

A.2.1 Raw Team Efficiency

The three plants shared detailed daily output data for each worker and task, the standard allowable time (SAT) of each task, and the actual seconds worked by each worker per task and day. We calculate daily team efficiency for team j on day d as

$$\text{Eff}_{jd} = \frac{\sum_{i=1}^{n_{jd}} \sum_{o=1}^{k_i} \text{output}_{io} \times \text{SAT}_o}{\sum_{i=1}^{n_{jd}} H_{ijd}}, \tag{2}$$

where n_{jd} is the number of workers on team j on day d, k_i is the number of tasks performed by worker i, output_{io} is worker i's completed output of task o, SAM_o is the standard allowed minutes for task o, and H_{ijd} is the actual seconds worked by worker i on team j on day d. The numerator is thus the total standard time of all tasks the team completed in a day, and the denominator is the actual time the team spent working; their ratio is the team's realized output relative to the benchmark. Weekly efficiency is the mean of daily efficiency over the week. This construction is the standard measure of line/team efficiency used in the garment manufacturing industry.

Raw efficiency contains variation that is unrelated to treatment and adds noise to the estimation. Equation 2 measures a team's realized output relative to a benchmark, but does not account for the fact that efficiency across teams may not always be comparable: some are producing more complex styles than others, and some have had more days of practice on their current style than others. Because treatment is randomized, there is no reason to expect these factors to be correlated with treatment status in expectation, so raw efficiency comparisons across arms are not biased by them. However, style and tenure differences still generate team-to-team variation in raw efficiency that has nothing to do with the treatment. With a modest number of clusters (42 teams), this nuisance variation inflates the standard errors of the treatment-effect estimate and reduces statistical power.

A.2.2 Team Efficiency Residualized on Style Fixed Effects and the Learning Rate

Raw team efficiency (equation 2) bundles together two sources of variation that are unrelated to the intervention itself: (i) which garment style a team happens to be producing in a given period, since styles differ in production difficulty, and (ii) how long the team has already been producing that style, since efficiency rises mechanically with practice (learning-by-doing). To remove both sources of variation, we estimate, separately for each plant p,

$$\text{eff}_{jd} = \alpha_p + \lambda_s + \theta \ln(\text{tenure}_{jd}) + u_{jd}, \tag{3}$$

where λ_s is a fixed effect for the style s that team j is producing on day d, and tenure_{jd} is the team's cumulative number of production days on that day's style. The adjusted efficiency measure is the regression residual, re-centered on the sample mean

so that it remains on the same scale as raw efficiency:

$$\widetilde{Y}_{jd} = \hat{u}_{jd} + \overline{\text{eff}}. \tag{4}$$

Weekly values are the team-week means of the adjusted daily measure $\widetilde{Y}_{jd}$.

We also report a variant that residualizes on style fixed effects alone, dropping the $\theta \ln(\text{tenure}_{jd})$ term from equation 3; this isolates the contribution of style/complexity without also purging learning-by-doing, since treatment can affect learning rate.

Treatment effects on this measure are estimated with a post-treatment weekly panel ANCOVA,

$$Y_{jt} = \alpha + \beta_1 T1_j + \beta_2 T2_j + \rho\, \overline{Y}_j^{\text{pre}} + \delta_s + \gamma_t + \varepsilon_{jt}, \tag{5}$$

where $\overline{Y}_j^{\text{pre}}$ is the team's mean of the same measure over its pre-intervention weeks, δ_s denotes randomization batch $\times$ stratum fixed effects, and γ_t denotes calendar-week fixed effects.

Appendix Table A1.5 shows the regression results. The results are robust to controlling for style FE and learning. With style fixed effects alone, the intervention increases team efficiency by 0.057 (by 8.8%) for the efficiency information treatment and 0.071 (by 11.1%) for the efficiency and preference information treatment, both statistically significant. Controlling additionally for log style tenure yields similar estimates of 0.059 (by 9.4%) and 0.056 (by 8.9%), respectively, both significant at the 5% level. The treatment effects are not statistically different across the two treatment arms in either specification (p-values = 0.580 and 0.889).

A.2.3 Complexity-Adjusted Team Efficiency

Some styles are produced by only a handful of teams, so a full set of style dummies can partly absorb team-level fixed effects rather than purely capturing style complexity, which risks netting out genuine cross-team (and potentially treatment-related) variation along with the nuisance complexity variation.

To avoid this issue, we construct an alternative adjustment that nets out product complexity using the number of distinct tasks a garment style requires, rather than a full set of style dummies. Using only pre-intervention team-days, we estimate, separately by plant,

$$\text{efficiency}_{jt} = \alpha + \beta_1\, \text{ntasks}_{jt} + \theta \log(\text{tenure}_{jt}) + u_{jt}, \qquad t \in \text{pre-intervention}, \tag{6}$$

where ntasks_{jt} is the number of distinct operations required by team j's style on day t, a continuous proxy for product complexity. Because this regression is estimated only on pre-intervention data, the complexity adjustment cannot be contaminated by any treatment-induced change in efficiency or learning speed that occurs after the intervention begins.

We then construct the complexity-adjusted daily efficiency by subtracting only the estimated complexity component from raw efficiency:

$$\text{cadj_efficiency}_{jt} = \text{efficiency}_{jt} - \hat{\beta}_1 \, \text{ntasks}_{jt}. \tag{7}$$

Note that only the complexity component is removed; the learning term $\theta \log(\text{tenure}_{jt})$ is *not* subtracted, so any treatment effect that operates through the learning curve remains in the outcome. Finally, we aggregate to the team-week level:

$$\text{cadj_efficiency}_{jw} = \frac{1}{D_{jw}} \sum_{t \in D_{jw}} \text{cadj_efficiency}_{jt}, \tag{8}$$

where D_{jw} is the set of production days for team j in week w.

A.3 Construction of Demand-Adjusted Match Scores

For each team-day (j, t), workers are assigned to operation cells c, where a cell is a task category (a machine–complexity-level pair), a machine type, or a complexity level, depending on the outcome family f. For the multi-worker task allocation family there is no cell structure, and the score below is defined directly at the team-day level. Let i index workers. Each worker-task observation carries a percentile rank $\rho_i \in [0, 1]$ of the worker's performance, measured relative to a comparison group of size N_{jc} drawn from the team's roster of qualified workers, with higher percentiles denoting better performance.

A.3.1 Actual match score.

Let K_{jtc} denote the number of workers assigned to cell c on team-day (j, t). The cell-level actual match score is

$$\bar{a}_{jtc} = \frac{1}{K_{jtc}} \sum_{i=1}^{K_{jtc}} \rho_i. \tag{9}$$

A.3.2 Best-feasible benchmark.

The benchmark b_{jtc} is the highest average percentile attainable in cell c given the day's demand (K_{jtc} slots) and the team's roster of N_{jc} qualified workers. If percentiles are assigned by rank, $\rho(\text{rank} = r) = \frac{N_{jc}-r}{N_{jc}-1}$, with $r = 1$ denoting the best-performing worker in the comparison group, the highest attainable average percentile among any K_{jtc} of the N_{jc} candidates is the mean percentile of the top K_{jtc} ranks:

$$b_{jtc} = \frac{1}{K_{jtc}} \sum_{r=1}^{K_{jtc}} \frac{N_{jc} - r}{N_{jc} - 1} = \frac{N_{jc} - \frac{K_{jtc}+1}{2}}{N_{jc} - 1}. \tag{10}$$

A.3.3 Cell gap.

The cell gap is

$$g_{jtc} = \bar{a}_{jtc} - b_{jtc} \leq 0, \tag{11}$$

non-positive by construction, and measures the shortfall of the realized allocation from the theoretically best allocation attainable that day given the cell's demand and the team's worker pool.

A.3.4 Aggregation to the team-day level.

For the six cell-structured outcome families — efficiency, learning, variability, task preference, machine preference, and level preference — cell gaps are aggregated into a single team-day match score by weighting each cell by the worker-hours it absorbed that day. Let τ_{ijtc} denote worker i's actual time on cell c on team-day (j, t), and let $\omega_{jtc} = \sum_i \tau_{ijtc}$, summed only over workers whose match status is observed, so that each cell's time weight is consistent with the population over which g_{jtc} is computed. The team-day demand-adjusted match score for outcome family f is

$$\text{Gap}^f_{jt} = \frac{\sum_c \omega_{jtc}\, g_{jtc}}{\sum_c \omega_{jtc}}. \tag{12}$$

For the multi-worker task allocation family, the gap is defined directly at the team-

day level by applying equations (9)–(11) with K_{jt} and N_{jt} denoting team-day demand and supply.